



APXGP

HUGO ANDRÉS
RAMIREZ TOVAR

FORMULA 1 AND THE ART OF AERODYNAMICS



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	
1	results	year	round	circuit	Track	No of Driv	Drivers	surname	construct	Team	grid	position	positionTe	positionO	points	laps	time	millisecor	fastestLaç	rank	fastestLaç	fastestLaç	sta
2	1	2024	1	3 Bahrain		1	Max Verstap	9 Red Bull			1	1	1	1	26	57	31:44.7	5504742	39	1	01:32.6	210.383	
3	2	2024	1	3 Bahrain		11	Sergio Perez	9 Red Bull			5	2	2	2	18	57	22.457	5527199	40	4	01:34.4	206.468	
4	3	2024	1	3 Bahrain		55	Carlos Sainz	6 Ferrari			4	3	3	3	15	57	25.11	5529852	44	6	01:34.5	206.156	
5	4	2024	1	3 Bahrain		16	Charles Leclerc	6 Ferrari			2	4	4	4	12	57	39.669	5544411	36	2	01:34.1	207.069	
6	5	2024	1	3 Bahrain		63	George Russell	131 Mercedes			3	5	5	5	10	57	46.788	5551530	40	12	01:35.1	204.946	
7	6	2024	1	3 Bahrain		4	Lando Norris	1 McLaren			7	6	6	6	8	57	48.458	5553200	1	5	01:34.5	206.223	
8	7	2024	1	3 Bahrain		44	Lewis Hamilton	131 Mercedes			9	7	7	7	6	57	50.324	5555066	39	7	01:34.7	205.688	
9	8	2024	1	3 Bahrain		81	Oscar Piastri	1 McLaren			8	8	8	8	4	57	56.082	5560824	1	11	01:35.0	205.123	
0	9	2024	1	3 Bahrain		14	Fernando Alonso	117 Aston Mar			6	9	9	9	2	57	74.887	5579629	48	3	01:34.2	206.83	
1	10	2024	1	3 Bahrain		18	Lance Stroll	117 Aston Mar			12	10	10	10	1	57	93.216	5597958	30	16	01:35.6	203.73	
2	11	2024	1	3 Bahrain		24	Guanyu Zhou	15 Sauber			17	11	11	11	0	56	\N	\N	30	14	01:35.5	204.102	
3	12	2024	1	3 Bahrain		20	Kevin Magnusse	210 Haas F1Ti			15	12	12	12	0	56	\N	\N	34	15	01:35.6	203.863	
4	13	2024	1	3 Bahrain		3	Daniel Ricciardo	215 RB F1Tea			14	13	13	13	0	56	\N	\N	37	13	01:35.2	204.735	
5	14	2024	1	3 Bahrain		22	Yuki Tsunoda	215 RB F1Tea			11	14	14	14	0	56	\N	\N	37	18	01:35.8	203.303	
6	15	2024	1	3 Bahrain		23	Alexander Albon	3 Williams			13	15	15	15	0	56	\N	\N	40	17	01:35.7	203.537	
7	16	2024	1	3 Bahrain		27	Nico Haulkenb	210 Haas F1Ti			10	16	16	16	0	56	\N	\N	46	10	01:34.8	205.445	
8	17	2024	1	3 Bahrain		31	Esteban Ocon	214 Alpine F1			19	17	17	17	0	56	\N	\N	34	20	01:36.2	202.473	
9	18	2024	1	3 Bahrain		10	Pierre Gasly	214 Alpine F1			20	18	18	18	0	56	\N	\N	45	9	01:34.8	205.508	
0	19	2024	1	3 Bahrain		77	Valtteri Bottas	15 Sauber			16	19	19	19	0	56	\N	\N	33	19	01:36.2	202.523	
1	20	2024	1	3 Bahrain		2	Logan Sargeant	3 Williams			18	20	20	20	0	55	\N	\N	42	8	01:34.7	205.659	
2	21	2024	2	77 Saudi Arat		1	Max Verstap	9 Red Bull			1	1	1	1	25	50	20:43.3	4843273	50	3	01:31.8	242.188	
3	22	2024	2	77 Saudi Arat		11	Sergio Perez	9 Red Bull			3	2	2	2	18	50	13.643	4856916	37	8	01:32.3	240.876	
4	23	2024	2	77 Saudi Arat		16	Charles Leclerc	6 Ferrari			2	3	3	3	16	50	18.639	4861912	50	1	01:31.6	242.561	
5	24	2024	2	77 Saudi Arat		81	Oscar Piastri	1 McLaren			5	4	4	4	12	50	32.007	4875280	1	10	01:32.3	240.779	

DATASET

The project integrates data related to race results, team, driver, position, and time, with a wind tunnel summary.

Dimensions: F1_race_traces.csv (23 variables, 878 observations) and constructors_wind_tunnel.csv (6 variables, 10 observations).

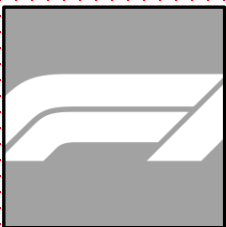
The project information was compartmentalized across multiple files, so I needed to integrate it, remove irrelevant data, and synthesize points earned by drivers and teams to provide evidence to support the project's conclusions.



Formula 1 - the power of data analysis, given that the difference between victory and defeat is measured in thousandths of a second.



During a race weekend, F1 teams employ between 30 and 50 data scientists to analyze and process information,



F1 car generates up to 300 GB of data per race weekend, and up to 200 different parameters are optimized.

**INTERESTING
STATISTICS OR
GROUPINGS**



Max Verstappen lap of 1.18.792 during the 2025 Italian grand prix qualifying has become the fastest lap done in F1 history



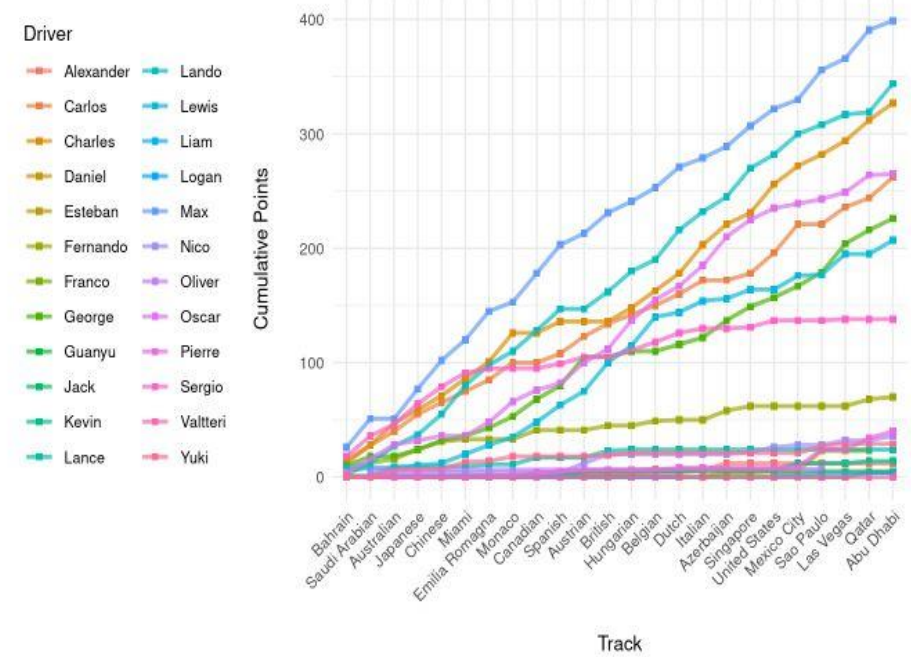
The fastest pitstop of the 2025 F1 season was 1.9 seconds by the McLaren F1 team on the Hungarian grand prix



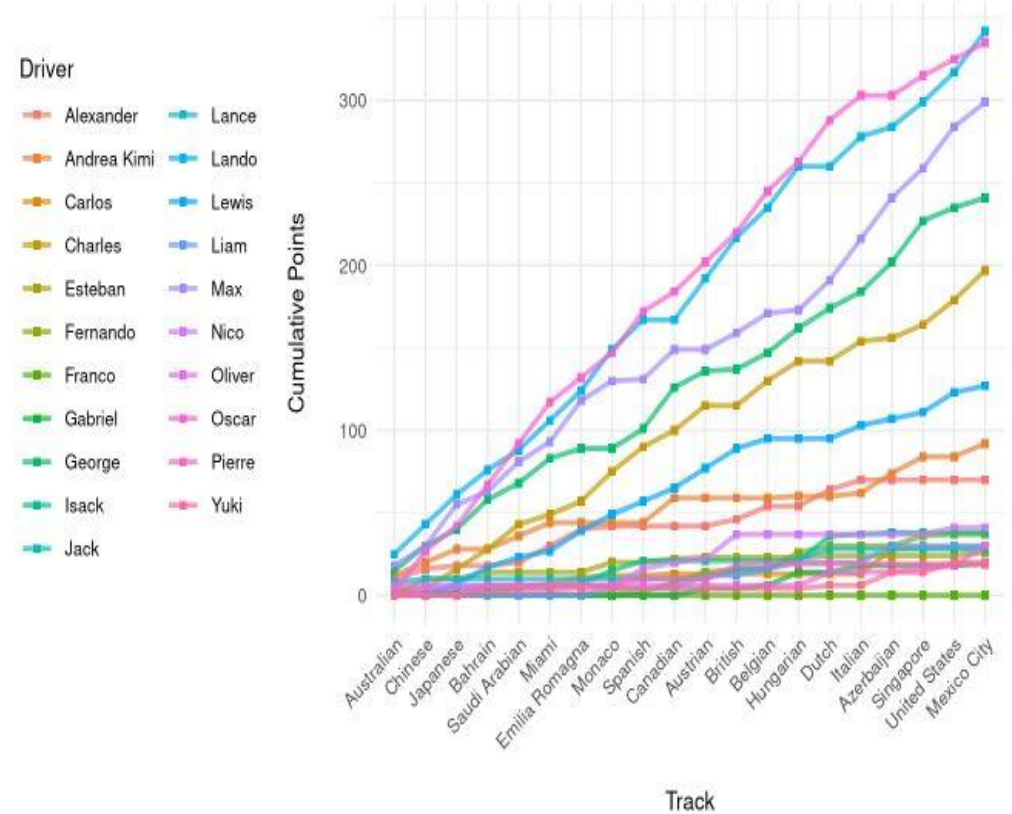
Andrea Kimi Antonelli podium on the Canadian grand prix converted him on the 3rd youngest driver to achieve one in F1 history

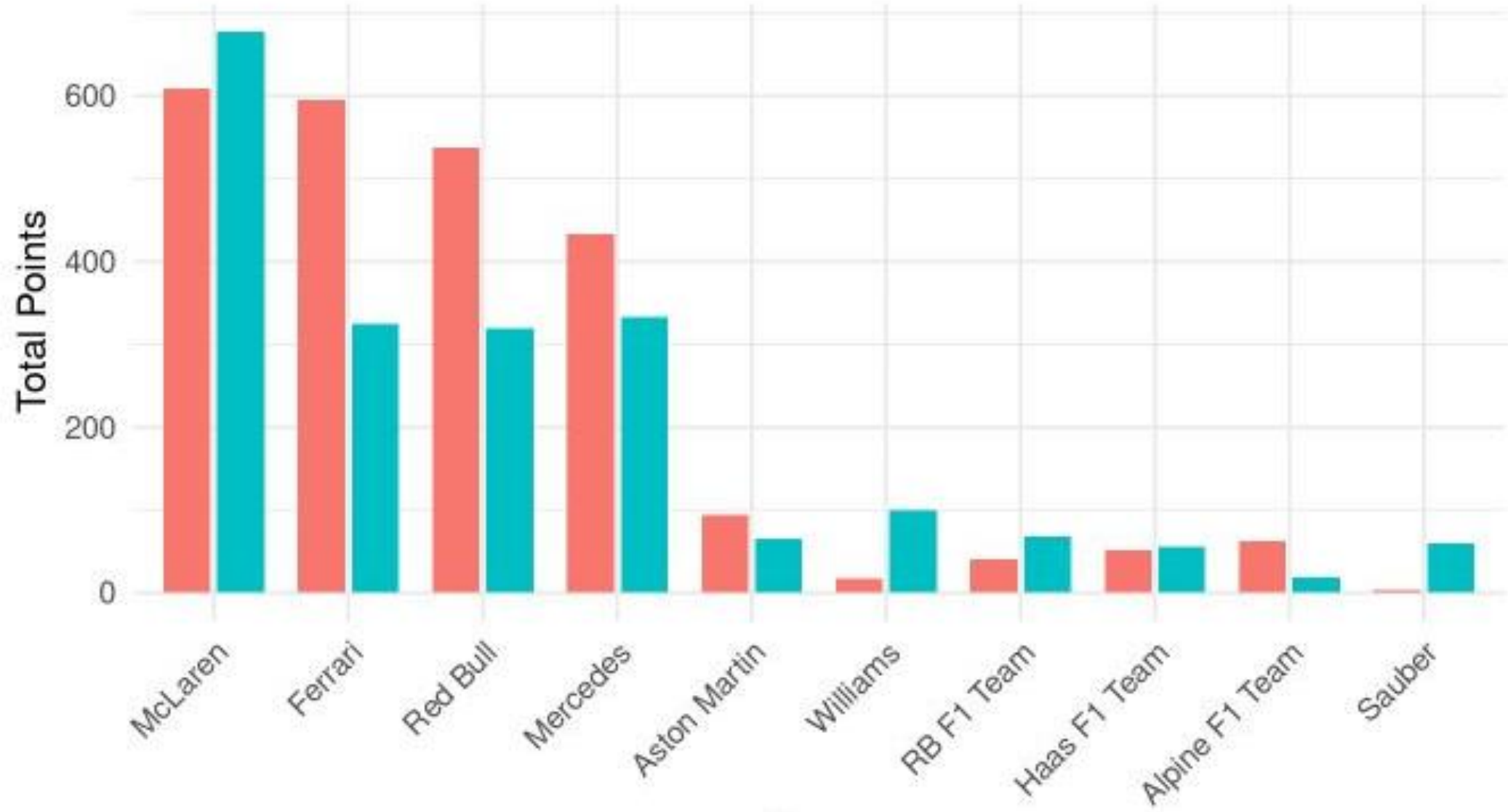
**INTERESTING
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2024 F1 Championship: Driver Points Evolution

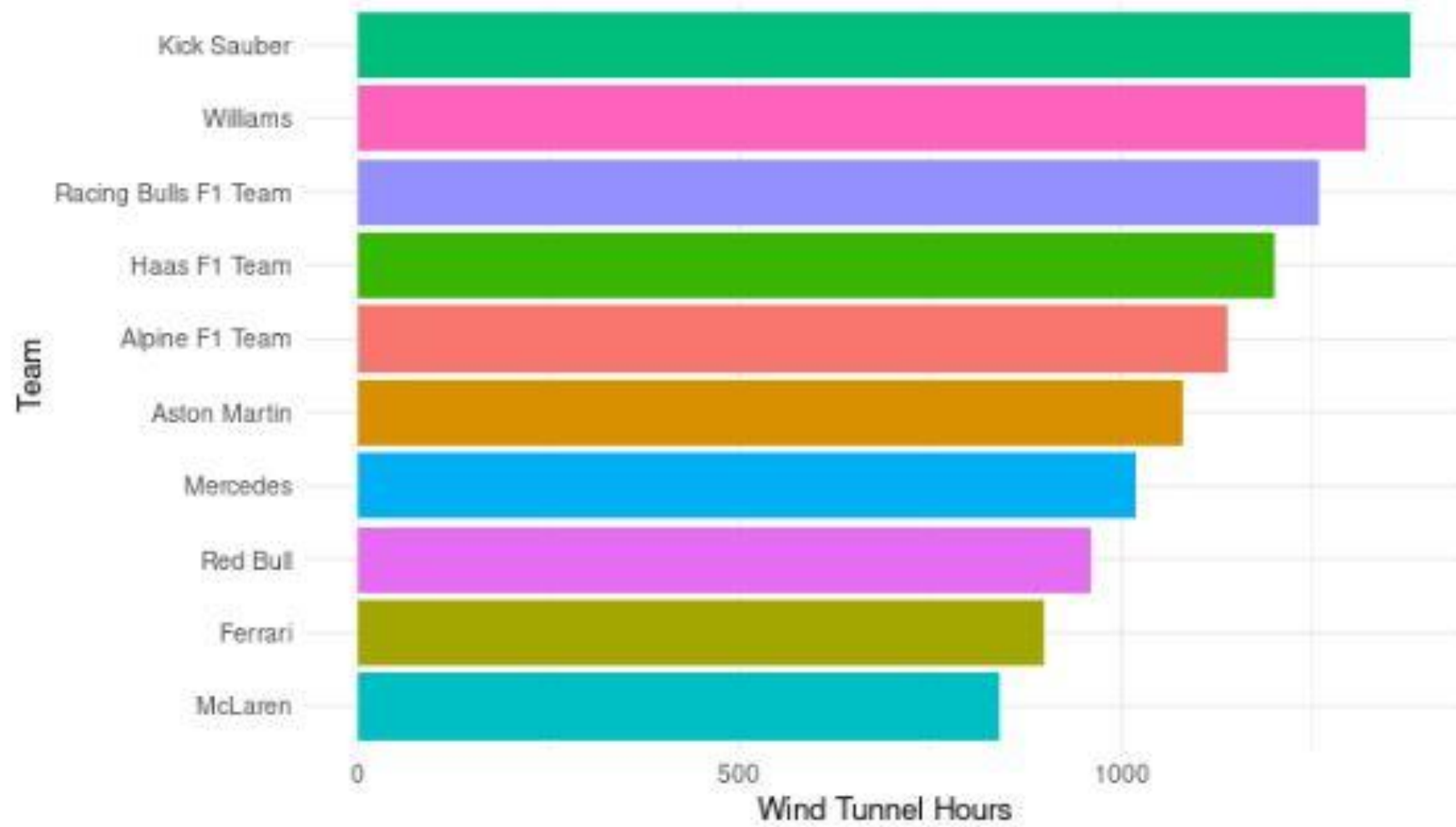


2025 F1 Championship: Driver Points Evolution

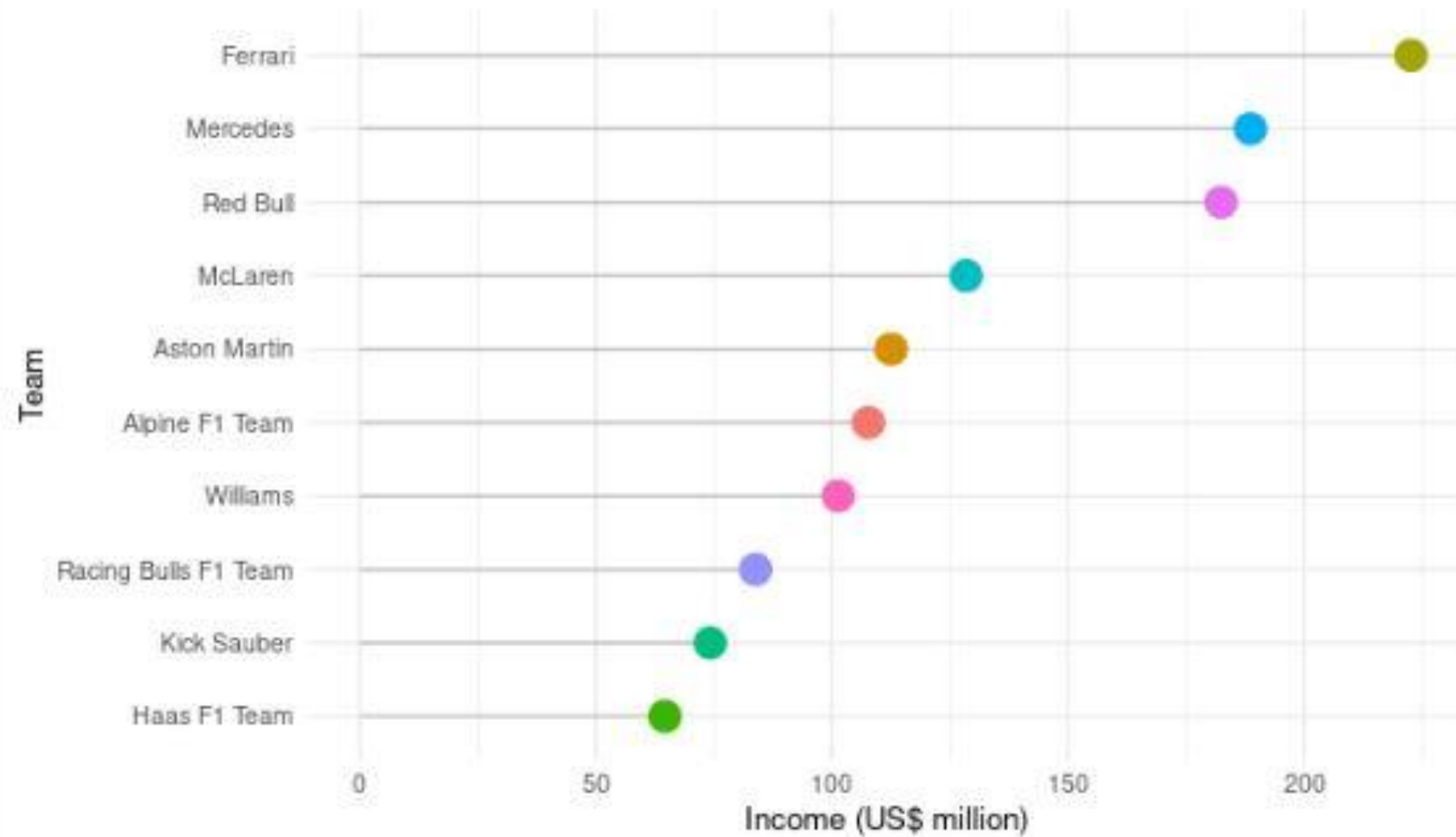




Wind Tunnel Hours by F1 Team (2025 Season)



F1 Team Income at End of 2024 Season



RESULTS



The project results show how teams that were near the bottom of the constructors' championship in 2024 are now consistently fighting for points in every race in the 2025 season.

In the 2024 season, the Williams team scored a total of 17 points, in only 6 out of 24 races. In the current 2025 season, the Williams team has scored a total of 111 points in 14 races, in addition to achieving a podium finish at the Azerbaijan Grand Prix.

The project suggests that this turnaround is due to the increased number of wind tunnel hours available for further aerodynamic development of their car. The same concept applies to teams like Haas and Kick Sauber, who have seen their performance improve thanks to the new FIA regulations.





F1
THE MOVIE
ONLY IN THEATERS
SUMMER 2025



CANADA EDITION

THANKS FOR YOUR TIME

APXCP

HUGO ANDRÉS
RAMÍREZ TOVAR

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